

Measuring Barriers to Structural Transformation: A Within-Country Perspective from Mexico

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This paper: Barriers to Mobility

- In Mexico, large wage gaps exist between agricultural and non-agricultural workers. Common interpretation: **barriers to sectoral mobility**.
- What can explain the magnitude of the observed barriers and how they differ across states in Mexico?
 - **Worker selection** on unobserved ability. More skilled workers allocate in non-agricultural sectors
 - **Technology** differences between sectors that transform human capital into productivity and therefore wages
- I use **individual panel data** from the ENOE labor survey in Mexico to distinguish between these potential explanations and explore the geographical heterogeneity Data: ENOE

Three main contributions

- Structural transformation literature usually uses cross-country aggregated data. I use a **within-country individual panel** from Mexico.
 - Institutions, legal/monetary regimes are held constant. **Subnational heterogeneity** becomes visible.
 - Model-implied estimates can be **validated** against observed sector switchers in the same dataset.
- Magnitudes are more consistent with **worker selection**. Policy implications: e.g. should we subsidize rural-urban migration?
- Literature measures the size of barriers. Often silent on the content. I show systematic associations with three frictions often discussed: **financial access, road infrastructure, and weather disruptions**.

Background: Wage and Schooling Gaps

- Large gaps between agriculture and non-agriculture persist even after adjusting for human capital. By sector

$$\log w_{isrt} = \delta_{rt} \mathbb{1}\{\text{non-agric}\}_{isrt} + \gamma s_{isrt} + X'_{isrt} \beta + \varepsilon_{isrt}$$

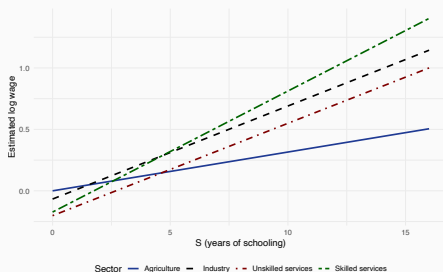
Estimated by state $\times$ year-quarter. **Wage gap** $\equiv \exp(\hat{\delta}_{rt})$.

	Wage gap $\exp(\hat{\delta})$	Schooling gap (yrs)
Raw	1.57 (0.09)	3.63 (0.26)
(+) Geog. + Gender	1.42 (0.06)	2.65 (0.37)
(++) Human Capital	1.22 (0.05)	—

Read: at the median, non-agricultural workers earn **22% more** than agricultural workers with the same schooling, experience, gender, and geography.

Background: Sectoral Mincer Returns Differ

- Classical frameworks assume a single Mincer return economy-wide. In Mexico, the **sectoral returns γ_j differ** with agriculture's slope flatter than other sectors.
- Different γ_j has direct implications for how we rationalize sectoral wage gaps as barriers to mobility.



Mexico City. Estimates from a log wage regression as before is reported $\widehat{\log w_j(s)} = \hat{\alpha}_j + \hat{\gamma}_j s$

Two Interpretations of Wage Gaps

- Herrendorf and Schoellman (2018) develop a partial equilibrium model that accounts for these features. Model
- Closed-form lower bounds for barriers under two interpretations:
 - **Selection view** ($\gamma_a = \gamma_n$): same technology, workers differ in unobserved ability.

$$\frac{1}{1 - \tau^{\min}} = \max \left\{ 1, \frac{W_n(0)}{W_a(0)} \right\} \quad (1)$$

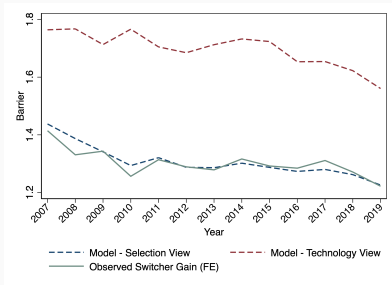
- **Technology view** ($\gamma_a < \gamma_n$): sectors differ in human-capital intensity.

$$\frac{1}{1 - \tau^{\min}} = \max \left\{ 1, \frac{W_n(\bar{s})}{W_a(\bar{s})} \right\} \quad (2)$$

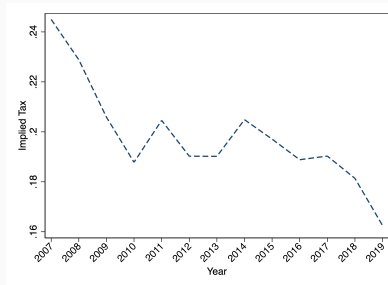
- Panel data on observed **sector switchers** serves as a tiebreaker between the two views.

Selection View Dominates

- Model-implied barriers under the selection view track the FE-based switcher estimates. The technology view produces implausibly large barriers. By state



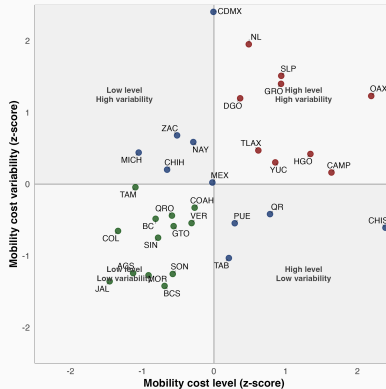
Panel a) National wage gaps



Panel b) National mobility cost τ

Mobility Costs: Levels and Variability

- States differ not only in the average level of mobility costs but also in variability over time.



Notes: Standardized measures centered on national moments. Horizontal axis: relative level of mobility costs. Vertical axis: relative temporal variability.

Opening the Black Box

- H&S estimate the **size** of barriers but are silent on what generates them. I provide suggestive evidence linking barrier variation to three frictions discussed in the literature. Decomposition
- **Financial access.** Development banking and ATM density together account for close to 10% of the variance. Commercial banking does not respond (Consistent with Aguilar-Gomez et al. (2024))
- **Road connectivity.** Changes in road density account for 5–6%.
- **Weather shocks.** About 2% of the variance, but consistent with a broader narrative linking adverse agricultural conditions to mobility frictions.

Conclusions

1. Selection view dominates within Mexico, across states and over time. Validated against individual-level panel data from the same dataset.
2. Mobility costs are substantial, heterogeneous across states, and volatile over time. Both dimensions are policy-relevant and vary spatially.
3. Financial access, road connectivity, and agricultural shocks are systematically associated with barrier variation. A first step toward opening the black box.
4. Limitations: correlates are suggestive, not causal. Variance decomposition accounts for a modest share. Further work needed on mechanisms.

Appendix

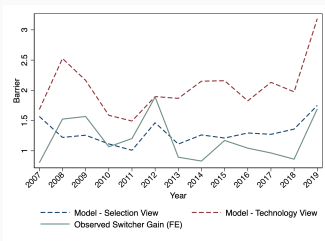
- National Survey of Occupation and Employment (ENOE), conducted by INEGI. Mexico's main labor force survey.
- Representative at the national and state level on a quarterly and yearly basis.
- Rotating panel design: households interviewed five times at three-month intervals. This generates an individual panel that allows tracking sectoral transitions.
- Covers on average about 550,000 unique households per year.
- Includes both formal and informal workers with detailed information on occupation, sector, wages, and demographics.
- Sample period: 2007–2019.

Wage and Schooling Gaps by Sector [Back](#)

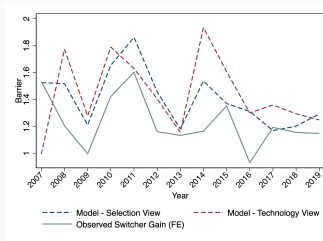
	Wage Gaps			Schooling Gaps		
	10th	Med.	90th	10th	Med.	90th
Non-agriculture						
Raw gap	1.30	1.57	2.28	2.89	3.63	4.32
(++) Human Capital	1.05	1.22	1.61			
Industry						
Raw gap	1.25	1.50	2.14	1.62	2.45	3.38
(++) Human Capital	1.06	1.26	1.64			
Unskilled services						
Raw gap	1.09	1.32	1.89	2.49	3.15	3.78
(++) Human Capital	0.91	1.07	1.40			
Skilled services						
Raw gap	1.59	1.94	2.97	3.97	5.00	6.02
(++) Human Capital	1.21	1.43	1.99			

Notes: Wage gaps as ratios relative to agriculture. Schooling gaps in years.

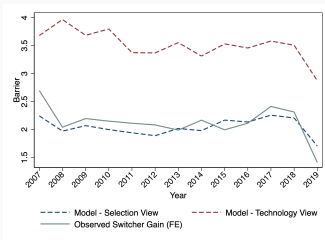
Mobility Barrier by State

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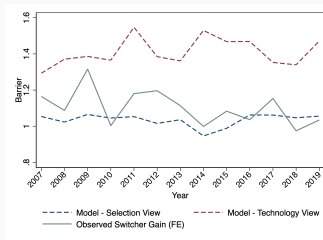
(a) Mexico City



(b) Nuevo Leon



(c) Chiapas



(d) Jalisco

Variance Decomposition [Back](#)

Model	Estimate [95% CI]	Share expl.	R^2
Panel A. Financial access			
Baseline (FE only)	—	—	0.843
+ Dev. bank	−0.747 [−1.340, −0.155]	0.021	0.846
+ ATMs	−0.020 [−0.030, −0.011]	0.063	0.853
+ Commercial bank	0.068 [−0.012, 0.148]	0.010	0.844
All three jointly	—	0.093	0.857
Dev. bank	−0.862 [−1.450, −0.274]		
ATMs	−0.021 [−0.031, −0.012]		
Commercial bank	0.010 [−0.071, 0.090]		
Panel B. Road density			
Baseline (FE only)	—	—	0.898
+ Road dens. (area)	−0.009 [−0.014, −0.003]	0.052	0.904
+ Road dens. (pop)	−0.004 [−0.006, −0.001]	0.064	0.905
Both jointly	—	0.072	0.906
Road dens. (area)	−0.004 [−0.012, 0.003]		
Road dens. (pop)	−0.003 [−0.005, 0.000]		
Panel C. Rainfall deficit			
Baseline (FE only)	—	—	0.817
+ Δ Rain Deficit	0.127 [0.039, 0.214]	0.021	0.820
A: 314 obs (2010–19); B: 189 obs (2014–19); C: 416 obs (2007–19)			

Share expl. = $1 - \text{RSS}_{\text{full}} / \text{RSS}_{\text{FE-only}}$. 95% CIs in brackets.

- One-period economy with two sectors j : agriculture (a) and non-agriculture (n).
- Each worker has innate unobserved ability $x \in [0, \bar{x}]$ and observed schooling $s \in [0, \bar{s}]$.
- Human capital: $h(x, s) = \exp(xs)$.
- Workers supply labor inelastically to the highest-wage sector, net of mobility cost $\tau_n \in [0, 1]$:

$$W_a(x, s) \geq (1 - \tau_n) W_n(x, s).$$

- Production: $Y_j = \int h(x, s)^{\gamma_j} m_j(x, s) dx ds$.
- Equilibrium:
 - Wages equal marginal products: $W_j(x, s) = P_j h(x, s)^{\gamma_j}$
 - Labor market clearing: $m(x, s) = m_a(x, s) + m_n(x, s)$

References

- Aguilar-Gomez, Sandra, Emilio Gutierrez, David Heres, David Jaume, and Martin Tobal**, “Thermal stress and financial distress: Extreme temperatures and firms’ loan defaults in Mexico,” *Journal of Development Economics*, May 2024, 168, 103246.
- Herrendorf, Berthold and Todd Schoellman**, “Wages, Human Capital, and Barriers to Structural Transformation,” *American Economic Journal: Macroeconomics*, April 2018, 10 (2), 1–23.